S&P / Platts Hydrogen Pricing

Original email to Santi was operating under a worst case scenario assumption of $0.06 / kWh. Since then, we have had conversation with PSO and OG&E to confirm that electrical pricing for companies creating renewables (including charging electric cars) in Oklahoma is $0.035 / kWh. The below numbers have been updated to reflect that and will be meeting with Santi to discuss.

S&P / Platts Hydrogen Standard Pricing Breakdown

- Delivered (within 50 miles, gaseous hydrogen at ~200 bar): $7.05/kg (30% profit margin)
- At-the-gate cost (excluding delivery, retaining 30% margin): $6.15/kg
- At-the-gate cost without R&D expenditures (30% margin): $4.36/kg

Our operational target to begin commercial hydrogen sales is Q3 2026.

Quick context about Tobe Energy:

- Proprietary electrolysis process with significant efficiency improvements.
- Fully American-made, vertically integrated manufacturing—from power electronics to electrolysis stacks, all in-house.

Feel free to attribute this information directly to Tobe Energy Corp. if desired. You're welcome to use or adapt the provided data below as it best fits your typical reporting structure:

PLATTS LOW-CARBON HYDROGEN: Renewable-derived gaseous hydrogen indicative value at $7.05/kg delivered within 50 miles (compressed, ~200 bar tube trailer), utilizing low-cost renewable electricity and sustainable water resources in Oklahoma. Price reflects a 30% profit margin at full commercial scale (~900,000 kg/year). At-the-gate offer, excluding delivery and ongoing R&D investments, is approximately **$4.36/kg.**

Details:

- **Location:** Oklahoma, USA
- **Production Method:** Electrolysis (90% system efficiency), renewable-derived low-carbon hydrogen
- **Delivery Method:** Gaseous hydrogen via tube trailers (~200 bar), delivered up to 50 miles
- **Price Basis:** Indicative delivered cost at scale, year-three facility maturity, 30% margin
- **Long-term Outlook:** Pricing expected to trend lower with further scale-up and improved operational efficiencies beyond year three
- **Sustainability Credentials:** Renewable energy sources, minimal environmental footprint, low-carbon hydrogen designation in alignment with Platts methodology

Table Breakdown of Worst-Case Scenario Cost for Tobe Energy

Item	Cost ($/kg)
Electricity (42 kWh/kg at $0.035/kWh)	$1.47
Compression & Purification (3.5 kWh/kg at $0.035/kWh)	$0.12
Transportation (50-mile delivery)	$0.90
RO Water (9 liters/kg at $0.005/liter)	$0.05
Direct OpEx (excluding CAPEX & labor)	$2.54
Annualized CAPEX & Personnel Costs (Including R&D capex and personnel requirements here)	$2.86
Total Breakeven Cost (Year 3)	$5.40
Indicative Selling Price (30% profit margin)	$7.05

Comparable Hydrogen Pricing from Platts

Conventional (Fossil Fuel) Hydrogen

Price ($/kg)	Form	Pickup terms	Volume cap	Expiry	Location	Source	Purity tag
7.2	Liquid (cryogenic)	Pickup only	≤ 1 trailer/day	16 May 2025	Charleston, TN	Plug Power	99.999 % (SAE 2719)
7.2	Liquid (cryogenic)	Pickup only	≤ 1 trailer/day	16 May 2025	Woodbine, GA	Plug Power	99.999 %
7.2	Liquid (cryogenic)	Pickup only	≤ 1 trailer/day	16 May 2025	St Gabriel, LA	Plug Power	99.999 %
9.9	Liquid (cryogenic)	Pickup only	≤ 1 trailer/day	28 Mar 2025	Woodbine, GA	Plug Power	99.999 %
9.9	Liquid (cryogenic)	Pickup only	≤ 1 trailer/day	28 Mar 2025	Charleston, TN	Plug Power	99.999 %
10.85	Liquid (cryogenic)	Pickup only	≤ 1 trailer/day	4 Apr 2025	Charleston, TN	Plug Power	99.999 %
10.85	Liquid (cryogenic)	Pickup only	≤ 1 trailer/day	4 Apr 2025	Woodbine, GA	Plug Power	99.999 %
13.9	Liquid (cryogenic)	Pickup only	≤ 1 trailer/day	11 Apr 2025	Woodbine, GA	Plug Power	99.999 %
13.9	Liquid (cryogenic)	Pickup only	≤ 1 trailer/day	11 Apr 2025	Charleston, TN	Plug Power	99.999 %

Green Hydrogen

Price ($/kg)	Form / pressure	Pickup / gate terms	Volume	Location	Timing	Notes	Purity tag
7.77	Gas, plant-gate	At-gate	~900 t/yr	Oklahoma	Q3 2026	Tobe Energy, renewables, ex-IRA	99.999 % (SAE 2719)
19	Gas, plant-gate	At-gate (take-or-pay)	10–17 t/yr (yrs 1-3)	California	5-yr contract (May 2024)	Transit sector	99.999 %
> 6	Not specified	At-gate	—	California	Developer quote	Renewables, ex-IRA	99.999 %
10–19	Not specified	At-gate	—	California	2030 target	Low-carbon LH_2, ex-IRA	99.999 %

Conventional Hydrogen (Out of Spec)

Price ($/kg)	Form & pressure	Pickup/gate terms	Volume note	Location	Remarks
8	Liquid (cryogenic)	Pickup at plant	< 5 t/mo	Southern CA	Spot, purity not stated
8–12	Liquid (cryogenic)	Facility gate	—	Southern CA	Spot range heard 3/28/25
9	Liquid (cryogenic)	At-gate	—	Nevada	SMR, purity not stated
9–11	Liquid (cryogenic)	At-gate	Spot	US Gulf Coast	Range, purity not stated
12	Gas (tube-trailer)	At-gate	< 5 t/mo	Southern CA	Spot
12.28	Gas (~200–300 bar)	Pickup at plant	< 5 t/mo	Southern CA	Spot
13	Liquid (cryogenic)	At-gate	< 5 t/mo	Southern CA	Spot

Methodology for Comparative Hydrogen-Price Benchmarking

To ground our cost claims in an objective market context we began with the complete set of S&P Global Commodity Insights / Platts hydrogen price "heards," bids, and spot offers listed in the appendix. After de-duplicating repeated entries and standardising the formatting (currency, units, and spelling—e.g., correcting "Toby" to Tobe Energy), each price point was tagged along three axes:

- Supply type (grey SMR "Conventional," low-carbon, or renewable/green)
- Delivery mode (pick-up at plant gate, delivered by tube-trailer, dispensed at a public pump, or shipped by pipeline)
- Subsidy status (explicit mention of IRA 45V, LCFS, carbon credits, or other incentives)

With those tags in place we applied the same commercial filters that govern Tobe Energy's own offering. We kept only prices that:

- exclude any embedded subsidy;
- are quoted at the plant gate or for customer pick-up (no retail pump, no long-haul freight, no captive pipeline);
- already include liquefaction or ≥200 bar compression so downstream users incur no extra compression cost; and
- for "Conventional" hydrogen, explicitly state fuel-cell-grade 99.999 % purity if we want a perfect apples-to-apples comparator.

Any quote that satisfied every filter became Tier 1 Conventional—the precise incumbent product we intend to undercut. Offers that passed the economic tests but lacked one or two metadata points (for example, the purity spec was implied rather than stated, or the listing carried no expiry date) were relegated to Tier 2 Conventional. A small handful of unsubsidized renewable or low-carbon quotes that met the same delivery criteria were grouped under Tier G Renewable. All other entries—pump prices in the high-$20s, pipeline tariffs quoted in dollars per million SCF, forward cost curves for 2035 mega-projects, and any figure built on IRA or LCFS credits—were excluded from the comparative analysis because they do not represent costs our customers would face at the point of purchase.

This disciplined triage yields a market snapshot that aligns cleanly with Tobe Energy's Oklahoma economics: today's fully on-spec SMR hydrogen collected at the plant gate trades in the $7–14 /kg range; broader but slightly less documented SMR quotes cluster around $8–13 /kg; and truly green, unsubsidized alternatives remain scarce and generally higher-priced, with the notable exception of Tobe Energy's own $7.77 /kg at-gate indication. These tiers provide a transparent baseline against which our updated costs—$6.24 /kg at the gate (dropping to $4.45 /kg as R&D amortizes) and $7.14 /kg delivered within 50 miles—can be compared. In short, the methodology ensures that every number we cite is a like-for-like match on purity, compression, delivery responsibility, and subsidy treatment, giving investors and customers a clear view of the discount our technology unlocks.

Appendix: All Hydrogen Pricing From Platts

Transit

PLATTS HYDROGEN: Liquid hydrogen indicative value below $10/kg at the pump, transit agency with 50-100 hydrogen fuel cell electric buses, US

PLATTS HYDROGEN: Liquid hydrogen indicative value at $10/kg at the pump, transit agency with 20-30 hydrogen fuel cell electric buses, US

PLATTS LOW-CARBON HYDROGEN: Low-carbon liquid hydrogen indicative breakeven fuel value at $7/kg at the pump, transit agency, California

PLATTS LOW-CARBON HYDROGEN: Renewable derived ‘green’ hydrogen, liquid, indicative breakeven fuel value at or above $8/kg at the pump, transit agency, California

PLATTS HYDROGEN: Indicative transacted value at $10/kg at the pump, transit agencies in California, short term contracts

Low-carbon

PLATTS LOW-CARBON HYDROGEN: Renewable derived hydrogen 5-year contract heard at $19/kg for 5-year contract, done in May 2024, gaseous hydrogen at the gate, 3-year base term, maximum of 10,000/kg for first year, 15,000/kg for year 2 and 17,000/kg for year 3, respectfully, both take or pay, and year 4 and 5 to be negotiated at a later date, going into transit sector, California

PLATTS LOW-CARBON HYDROGEN: Low-carbon liquid hydrogen indicative target value at $5-7/kg at the pump, transit agency, California, 2030

PLATTS LOW-CARBON HYDROGEN: Renewable derived hydrogen indicative value at $14/kg, going into transit sector, California

PLATTS LOW-CARBON HYDROGEN: Renewable derived hydrogen indicated above $6/kg in California, at the gate: Developer

PLATTS HYDROGEN: Indicative value at $9/kg SMR derived (grey) hydrogen in Nevada, at the gate, liquid hydrogen

PLATTS HYDROGEN: Indicative value at $12/kg, SMR-derived hydrogen, at-the gate, spot basis, gaseous, below 5,000 kg/month, Southern California

Other:

PLATTS HYDROGEN: Liquid hydrogen indicative bid below $7/kg, at the facility, Lancaster California, 10-year contract

PLATTS HYDROGEN: Indicative maintenance costs at 20 cents/kg to $1/kg, transit sector in California, short term contract with onsite, industrial gas production

PLATTS HYDROGEN: Indicative spot value at $12/kg, SMR-derived hydrogen, Southern California, at-the-gate, gaseous, below 5,000 kg/month

PLATTS CONVENTIONAL HYDROGEN: Spot offer at $9.90/kg, liquid hydrogen, pickup only, expires March 28, up to 1 trailer/day, 99.999% minimum purity, grade SAE 2719 compliant, Location: Woodbine, Georgia, Source: Plug Power

PLATTS CONVENTIONAL HYDROGEN: Spot offer at $9.90/kg, liquid hydrogen, pickup only, expires March 28, up to 1 trailer/day, 99.999% minimum purity, grade SAE 2719 compliant, Location: Charleston, Tennessee, Source: Plug Power

- Hydrogen

PLATTS CONVENTIONAL HYDROGEN: Spot offer at $9.90/kg, liquid hydrogen, pickup only, expires March 28, up to 1 trailer/day, 99.999% minimum purity, grade SAE 2719 compliant, Location: Woodbine, Georgia, Source: Plug Power

PLATTS CONVENTIONAL HYDROGEN: Spot offer at $9.90/kg, liquid hydrogen, pickup only, expires March 28, up to 1 trailer/day, 99.999% minimum purity, grade SAE 2719 compliant, Location: Charleston, Tennessee, Source: Plug Power

PLATTS RENEWABLE HYDROGEN: Renewable-derived "green" gaseous hydrogen indicative value at $7.77/kg at the-gate, utilizing renewable electricity, excluding IRA 45V tax credits, Oklahoma, full commercial scale (~900,000 kg/year), operational date Q3 2026, Source: Toby Energy Corp.

PLATTS RENEWABLE HYDROGEN: Renewable-derived "green" gaseous hydrogen indicative value at $8.67/kg delivered within 50 miles (compressed, ~200 bar tube trailer), utilizing renewable electricity, excluding IRA 45V tax credits, Oklahoma, full commercial scale (~900,000 kg/year), operational date Q3 2026, Source: Toby Energy Corp.

PLATTS LOW-CARBON HYDROGEN: Renewable derived "green" hydrogen indicated breakeven value $12-$13/kg in California, delivered hydrogen, transit sector

PLATTS LOW-CARBON HYDROGEN: Renewable derived "green" hydrogen indicative bid at $14-15/kg, delivered hydrogen going into transit and trucking sector, gaseous or liquid, California hydrogen hub

PLATTS LOW-CARBON HYDROGEN: Renewable derived "green" hydrogen indicative value at $10/kg, going into transit sector, starting in 2027, California

PLATTS RENEWABLE-DERIVED HYDROGEN: Indian refinery tenders renewable hydrogen willingness to pay at $2.5/kg, Volume: 5,000 mt per annum, Tenure: 25 years

PLATTS RENEWABLE-DERIVED HYDROGEN: Indian refinery tenders renewable hydrogen Indicative offer at $5-6/kg, Volume: 5,000 mt per annum, Tenure: 25 years

PLATTS RENEWABLE HYDROGEN: 'green' hydrogen, indicative production value at $4-$12 per kilogram, in terms of oil equivalent indicated ranging from $200 to $600 per barrel, 2025, region/location not specified, Source: Saudi Aramco CEO

PLATTS RENEWABLE-DERIVED HYDROGEN: Renewable-derived hydrogen indicative production cost at $7.00/kg, expected operational start date 2030 onward, origin: Mexico

PLATTS RENEWABLE HYDROGEN: Renewable "green" hydrogen indicated value $3/kg in US Gulf Coast, 40 tons/day, including IRA 45V credits

PLATTS RENEWABLE HYDROGEN: Renewable-derived "green" hydrogen indicative offer at $5-$6/kg, 30/mt per day, fuel cell grade pure hydrogen, 500 bar pressure, solar and behind the meter renewables, Southern California

PLATTS RENEWABLE HYDROGEN: Behind the meter renewable "green" hydrogen indicated offer $3-$4/kg in US Gulf Coast, plant gate or delivered within 30 miles via pipeline, 40 tons/day, including IRA 45V credits

PLATTS HYDROGEN: Indicative value at $8-$15/kg, SMR-derived hydrogen, delivered, spot basis, US

PLATTS HYDROGEN: Indicative value at $5/kg, SMR-derived hydrogen, delivered, spot basis, liquid, US

PLATTS HYDROGEN: Indicative offer at $9/kg, SMR-derived hydrogen, delivered, spot basis, liquid, USGC

PLATTS HYDROGEN: Indicative value at $9-11/kg, SMR-derived hydrogen, at-the-gate, spot basis, liquid, USGC

PLATTS HYDROGEN: Indicative value above $1.00/kg, delivered SMR-derived hydrogen, pipeline, end use: fertilizers and refinery, USGC

PLATTS CONVENTIONAL HYDROGEN: Spot offer at $7.20/kg, liquid hydrogen, pickup only, expires May 16, up to 1 trailer/day, 99.999% minimum purity, grade SAE 2719 compliant, Location: Charleston, Tennessee, Source: Plug Power

PLATTS CONVENTIONAL HYDROGEN: Spot offer at $7.20/kg, liquid hydrogen, pickup only, expires May 16, up to 1 trailer/day, 99.999% minimum purity, grade SAE 2719 compliant, Location: Woodbine, Georgia, Source: Plug Power

PLATTS CONVENTIONAL HYDROGEN: Spot offer at $7.20/kg, liquid hydrogen, pickup only, expires May 16, up to 1 trailer/day, 99.999% minimum purity, grade SAE 2719 compliant, Location: St. Gabriel, Louisiana, Source: Plug Power

Hydrogen

PLATTS HYDROGEN: Indicative value at $0.90-1.00/kg, delivered SMR-derived hydrogen via pipeline, end use: refinery, contract base, 5-100 MMSCFD (millions of standard cubic feet per day) or 11,800-236,100 kg/day, USGC

PLATTS RENEWABLE-DERIVED HYDROGEN: [Heard after 1630 GMT, Mar. 26] Renewable-derived hydrogen indicative production cost at $3.00-4.00/kg, expected operational start date 2030 onward, origin: Colombia

PLATTS RENEWABLE-DERIVED HYDROGEN: [Heard after 1630 GMT, Mar. 26] Renewable-derived hydrogen indicative production cost at $6/kg, expected operational start date 2030 onward, origin: Colombia

PLATTS RENEWABLE-DERIVED HYDROGEN: [Heard after 1630 GMT, Mar. 26] Renewable-derived hydrogen indicative production cost at $3.50/kg, expected operational start date 2030 onward, origin: Colombia

PLATTS RENEWABLE-DERIVED HYDROGEN: [Heard after 1630 GMT, Mar. 26] Renewable-derived hydrogen indicative production cost preliminary estimated at $9/kg for a pilot project in the refining sector in Colombia

PLATTS RENEWABLE-DERIVED HYDROGEN: [Heard after 1630 GMT, Mar. 26] Indicative willingness to pay value (WTP) for renewable-derived hydrogen estimated at $2.4-3.0/kg for the refining sector in Colombia

PLATTS CONVENTIONAL HYDROGEN: Spot offer at $10.85/kg, liquid hydrogen, pickup only, expires April 4, up to 1 trailer/day, 99.999% minimum purity, grade SAE 2719 compliant, Location: Charleston, Tennessee, Source: Plug Power

PLATTS CONVENTIONAL HYDROGEN: Spot offer at $10.85/kg, liquid hydrogen, pickup only, expires April 4, up to 1 trailer/day, 99.999% minimum purity, grade SAE 2719 compliant, Location: Woodbine, Georgia, Source: Plug Power

PLATTS RENEWABLE-DERIVED HYDROGEN: Indian refinery tenders renewable hydrogen Indicative offer at $4.5-5/kg, Volume: 5,000 mt per annum, Tenure: 25 years

PLATTS RENEWABLE-DERIVED HYDROGEN: Indicative value for renewable hydrogen at $6/kg, Volume: 100 mt per annum, Tenure: 20 years

PLATTS RENEWABLE-DERIVED HYDROGEN: Indian refinery tenders renewable hydrogen Indicative offer at $6/kg, Volume: 5,000 mt per annum, Tenure: 25 years

PLATTS HYDROGEN: Hydrogen indicative value at $36/kg, public pump, Port of Oakland, California, Class 8 Fuel Cell Trucks, fuel origin: North Las Vegas via liquid hydrogen [Heard April 1]

PLATTS RENEWABLE HYDROGEN: Renewable-derived "green" hydrogen indicative value below $10/kg, below 30/mt per day, fuel cell grade pure hydrogen, up to 500 bar pressure, utilizing renewable electricity, Los Angeles, California, operational date 2028 [Heard April 1]

PLATTS HYDROGEN: 5 Iwatani California refueling stations heard selling at $29.99/kg: (Corrected)

PLATTS RENEWABLE-DERIVED HYDROGEN: USGC Indicative value at $3/kg renewable-based "green" hydrogen, at the gate/facility, with dedicated renewables, including IRA 45V tax credits, Source: Developer

PLATTS HYDROGEN: Indicative offer at $22/kg, SMR-derived hydrogen, delivered at the heavy-duty trucking station and including transportation costs, California [Heard 3/28/2025]

PLATTS LOW-CARBON HYDROGEN: Indicative willingness to pay value (WTP) at $5-6/kg, dispensed hydrogen at refueling station, heavy-duty trucking and transit, California [Heard 3/28/2025]

PLATTS HYDROGEN: Indicative value at $13/kg, SMR-derived hydrogen, at-the gate, spot basis, liquid, below 5,000 kg/month, Southern California [Heard 3/28/2025]

PLATTS HYDROGEN: Indicative value at $9-11/kg, SMR-derived hydrogen, at-the gate, spot basis, liquid, USGC [Heard 3/28/2025]

PLATTS RENEWABLE-DERIVED HYDROGEN: [Heard after 1630 GMT, April 3] Renewable-derived hydrogen indicative production cost at $4.00/kg, expected operational start date 2030 onward, origin: Colombia

PLATTS RENEWABLE-DERIVED HYDROGEN: [Heard after 1630 GMT, April 3] Renewable-derived hydrogen indicative production cost at $3.80-6.00/kg, expected operational start date 2030 onward, origin: Colombia

PLATTS RENEWABLE-DERIVED HYDROGEN: [Heard after 1630 GMT, April 3] Renewable-derived hydrogen indicative production cost at $3.50-5.00/kg, expected operational start date 2030 onward, origin: Mexico

PLATTS HYDROGEN: Indicative value at $8/kg, SMR-derived hydrogen, spot basis, liquid, below 5,000 kg/month, pickup at the plant, Southern California
PLATTS HYDROGEN: Indicative value at $12.28/kg, SMR-derived hydrogen, spot basis, gaseous, below 5,000 kg/month, pickup at the plant, Southern California

PLATTS RENEWABLE-DERIVED HYDROGEN: [Heard after 1630 GMT, April 4] Renewable-derived hydrogen indicative production cost at $7.50/kg, origin: Colombia

PLATTS CONVENTIONAL HYDROGEN: Spot offer at $13.90/kg, liquid hydrogen, pickup only, expires April 11, up to 1 trailer/day, 99.999% minimum purity, grade SAE 2719 compliant, Location: Woodbine, Georgia, Source: Plug Power

PLATTS CONVENTIONAL HYDROGEN: Spot offer at $13.90/kg, liquid hydrogen, pickup only, expires April 11, up to 1 trailer/day, 99.999% minimum purity, grade SAE 2719 compliant, Location: Charleston, Tennessee, Source: Plug Power

PLATTS HYDROGEN: Indicative value above $1.00/kg, delivered SMR-derived hydrogen, pipeline, end use: fertilizers, contract base 20-30 years, 200 MMSCFD (millions of standard cubic feet per day) 472,800 kg/day, USGC

PLATTS LOW-CARBON HYDROGEN: Biomass gasification-derived hydrogen indicated transaction value at $4.50/kg in the US, above 12 mt/day facility, going into Southern California, excluding IRA subsidies, carbon credits bundled

PLATTS LOW-CARBON HYDROGEN: Biomass gasification-derived hydrogen indicated cost at $1.91/kg in the US, above 12 mt/day facility, excluding IRA subsidies and carbon credits

PLATTS RENEWABLE HYDROGEN: Renewable-derived "green" hydrogen indicated above $5/kg in US Gulf Coast, including IRA 45V credits

PLATTS HYDROGEN: Indicative value above $1.00/kg, delivered SMR-derived hydrogen, pipeline, end use: fertilizers and refinery, USGC

PLATTS LOW-CARBON HYDROGEN: Indicative values for renewable-derived hydrogen in the range of $10-$19/kg, California, at the facility gate, excluding IRA 45V credits

PLATTS HYDROGEN: Indicative spot value at $8.50-12/kg, conventional SMR-based hydrogen, Southern California, at the facility gate